

Algorithm

Question: 02

01. Show that the lower bound of any comparison based sort is $\Omega(n \log n)$

02. Show that a heap can be constructed from an array in $O(n)$ time.

03. Show the operation of the Radix-sort on the following array.

$A = \{673, 424, 527, 639, 247, 826, 974, 428, 356, 652, 713\}$

04. Show the steps of sorting the following list of numbers using Quicksort. Show steps of each partition operation.

$A = \{15, 9, 5, 2, 3, 6\}$

05. Show how you can sort the numbers of the following min-heap A. Illustrate each step.

$A = \{1, 2, 3, 6, 8, 11, 13, 15, 17, 21, 23\}$.

06. Show the steps of the operation Extract-min() on the following min-heap.

$A = \{4, 1, 3, 2, 16, 9, 10, 14, 8, 7\}$

07. Show the steps of the operation Extract-max() on the following Max-heap.

$A = \{10, 9, 8, 7, 6, 5, 4, 3, 2, 1\}$